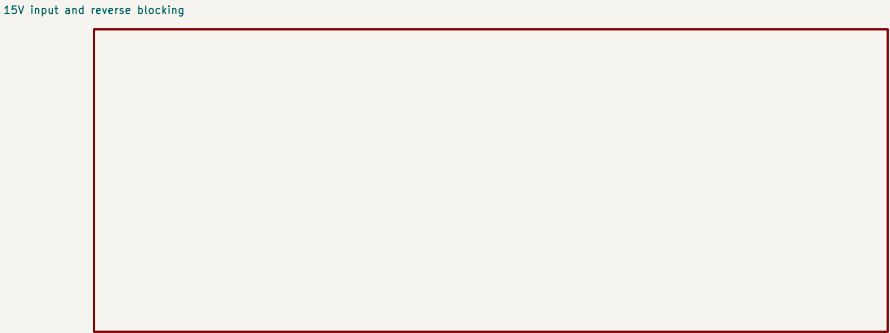
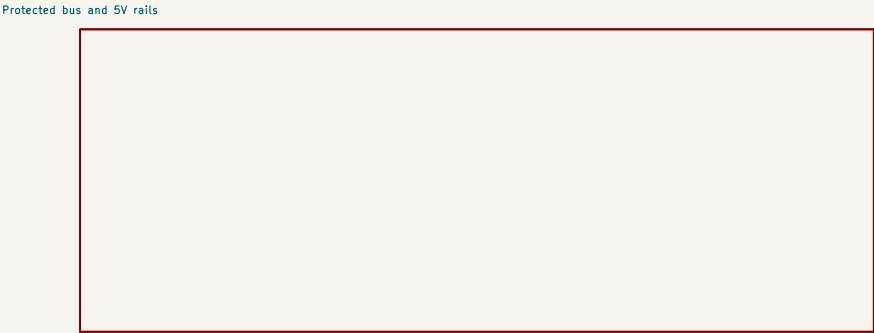


EDITABLE CIRCUIT CAPTURE C1 / not routed, fabricated or bench qualified

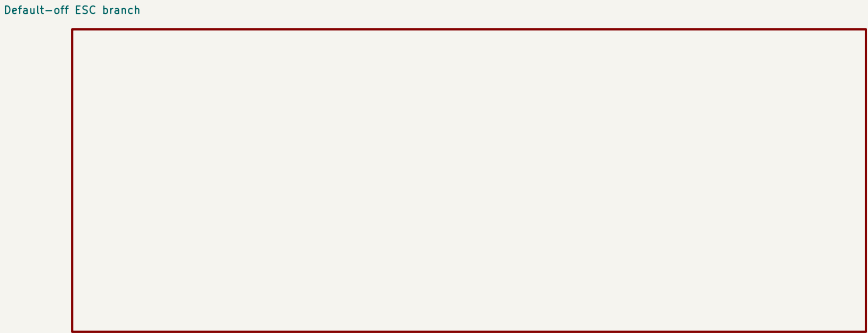
Every semiconductor pin, fitted passive and connector is explicit. NC pins have native no-connect flags.



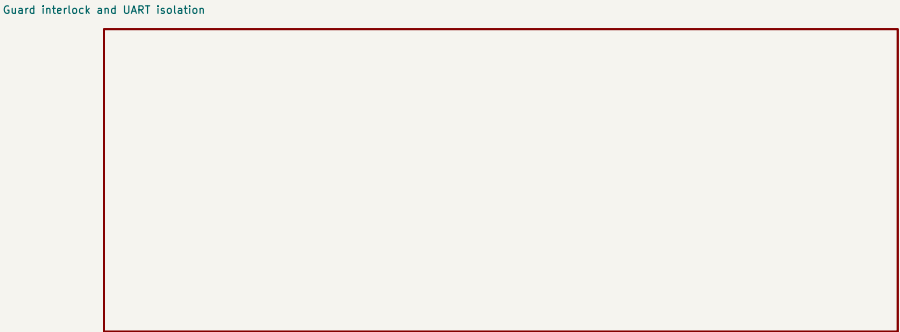
File: power.kicad_sch



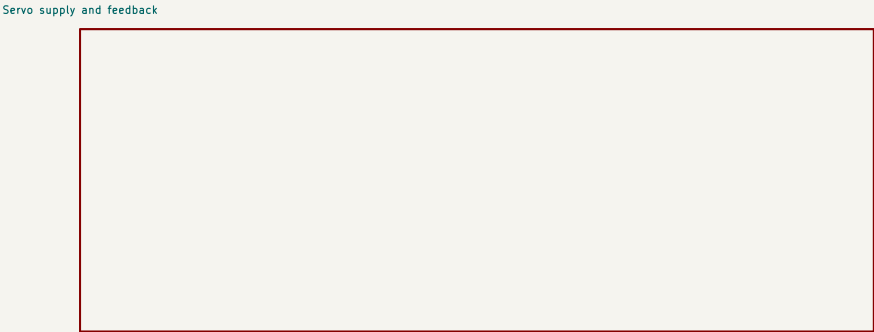
File: rails.kicad_sch



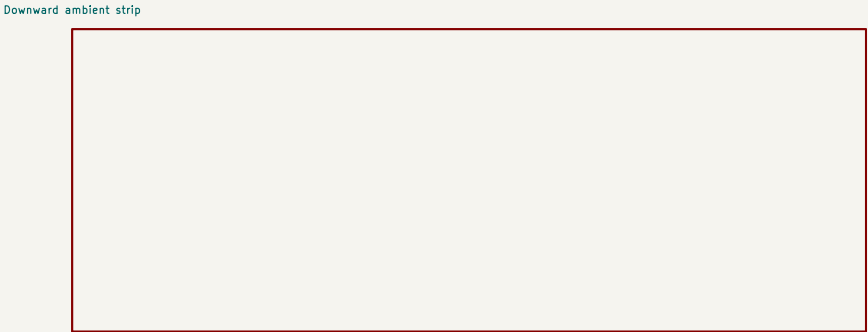
File: esc.kicad_sch



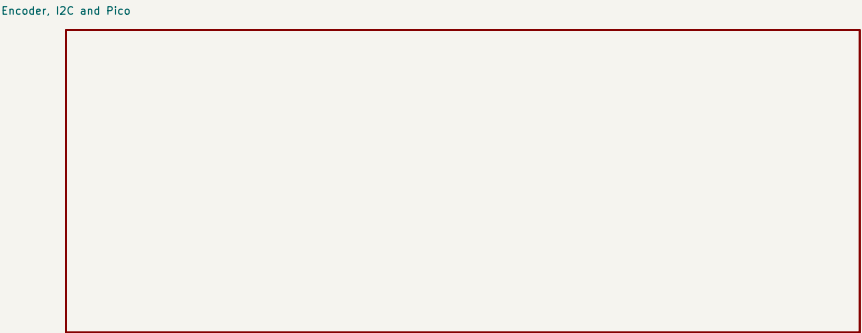
File: uart.kicad_sch



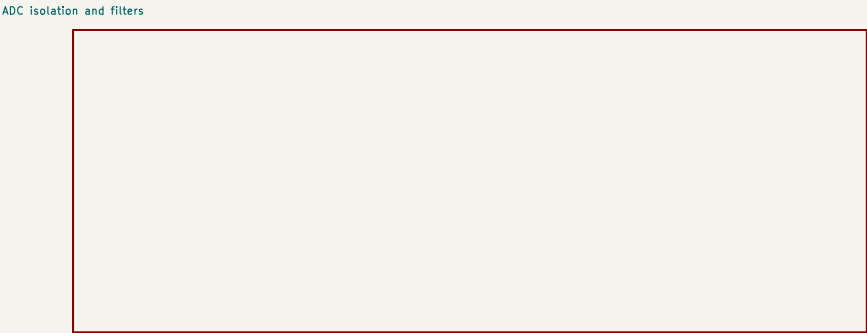
File: servo.kicad_sch



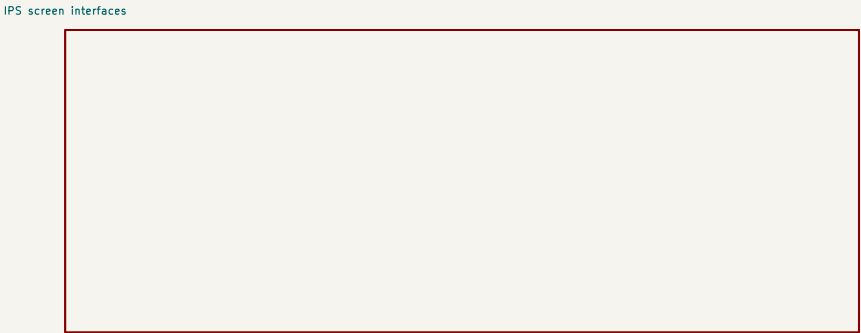
File: ambient.kicad_sch



File: inputs.kicad_sch

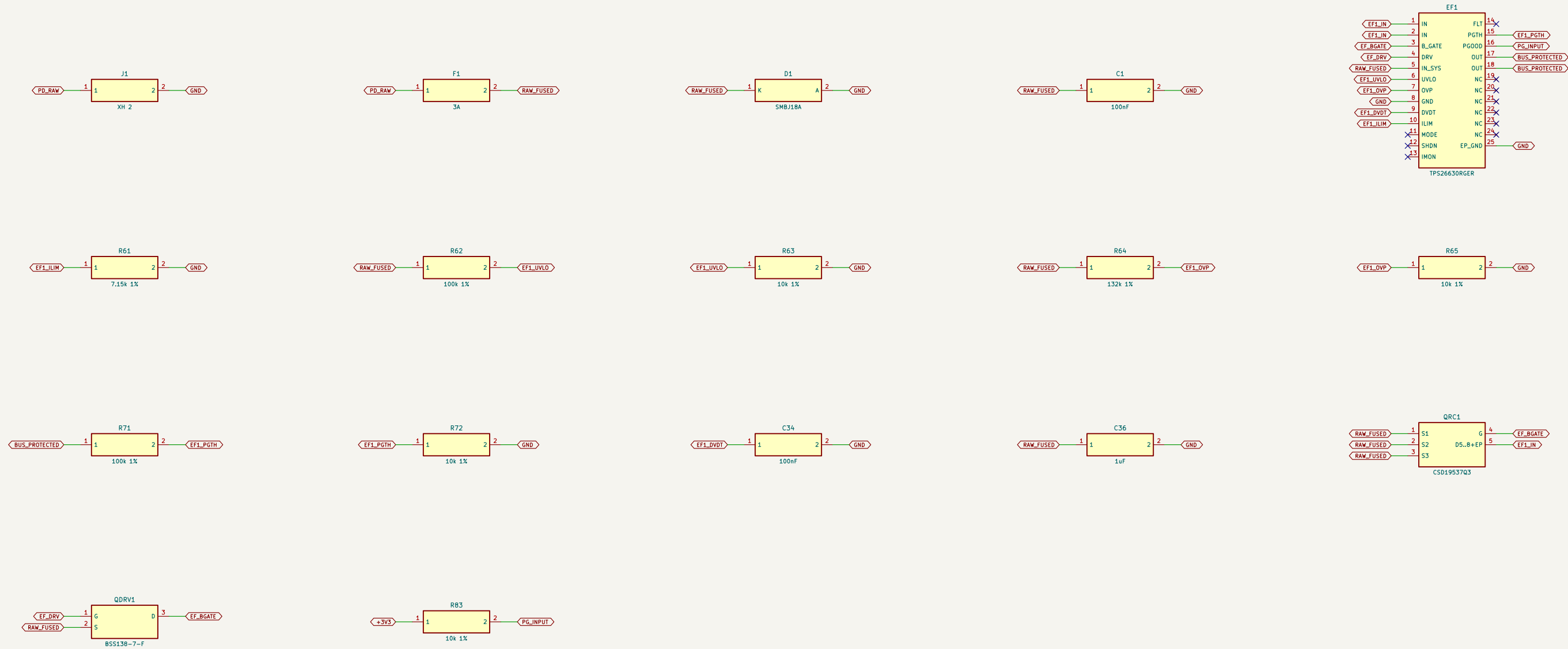


File: adc.kicad_sch



File: screen.kicad_sch

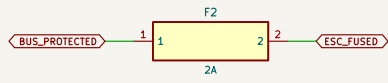
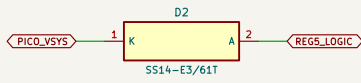
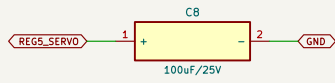
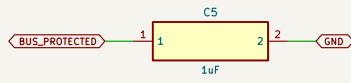
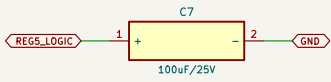
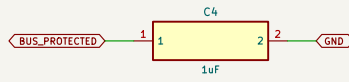
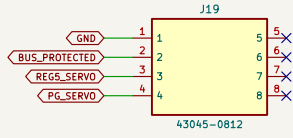
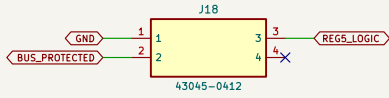
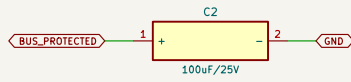




TPS26630 RGE has system GND on pin8 AND pad25; no isolated EF_RTN island.

QRC1 source=RAW_FUSED; drain=EF1_IN. QDRV1 gate=DRV, drain=BGATE, source=RAW_FUSED.

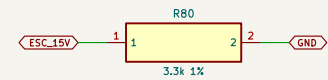
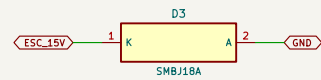
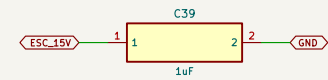
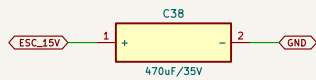
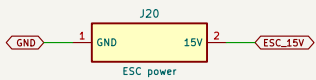
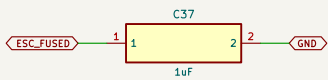
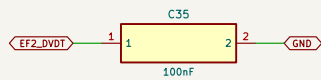
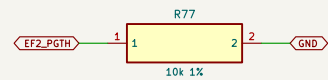
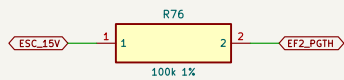
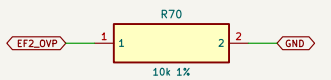
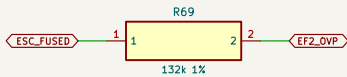
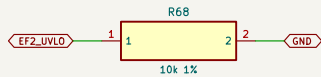
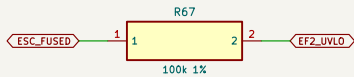
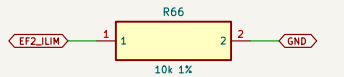
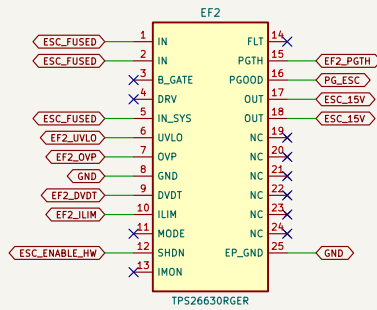
EF1 SHDN is intentionally open: specified open-circuit voltage2.48..3.3V. MODE open=latchoff.



15V3A PD contract required. No battery, no12V converter, no ESC BEC connection.

Remote D24V22F5 modules: input from protected bus, separate logic/servo5V outputs.

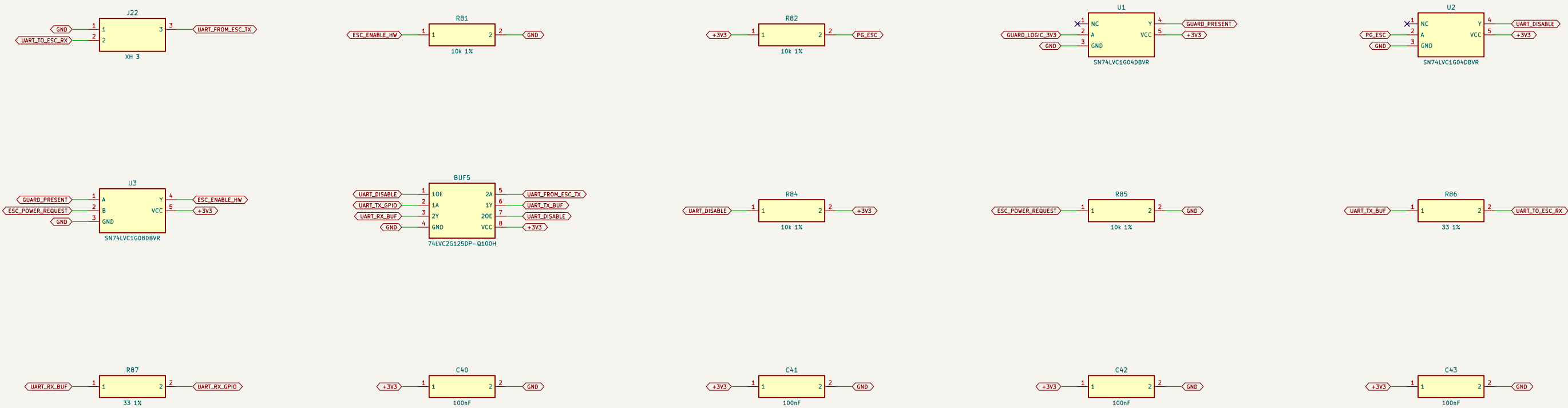
Regulator EN stays NC. PG_SERVO pullup is3.3V; modules must be qualified under load.



EF2 SHDN has10k pulldown. Guard AND MCU request enables it; reset/removal cuts motor power.

IN_SYS=IN=BUS_PROTECTED; B_GATE and DRV intentionally NC for this branch.

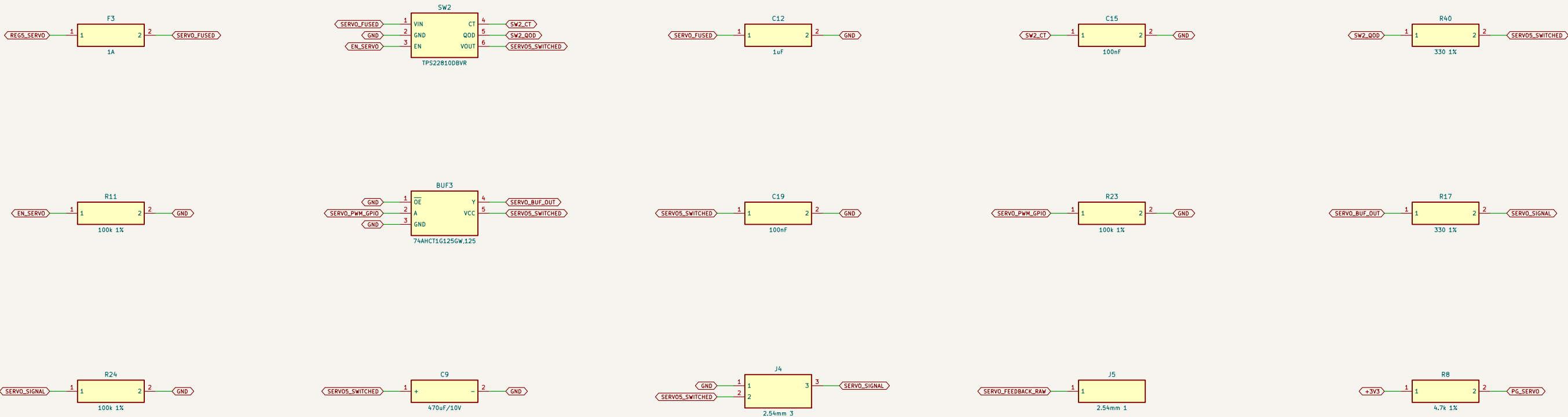
470uF35V, ceramic, TVS and3.3k bleed fit at ESC. No braking or regenerative energy sink.



BUF5 VCC=3.3V. Both /OE follow inverted ESC PG00D:10k pullup defaults disabled.

UART RX/TX are directionally buffered; max3.3V. ESC5V BEC remains physically insulated NC.

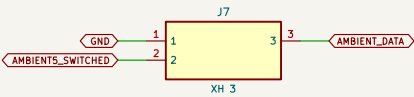
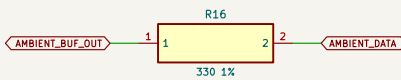
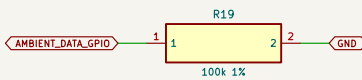
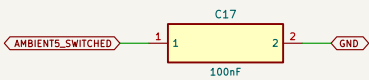
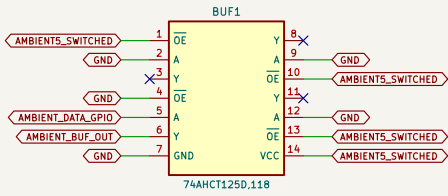
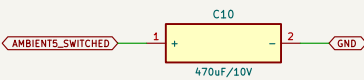
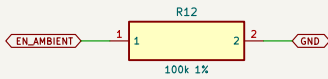
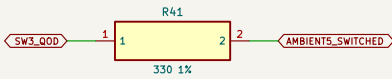
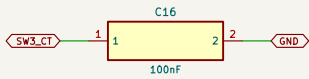
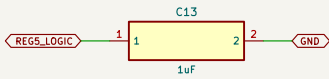
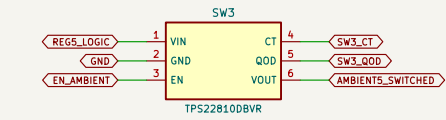
QRC1 symbol pad5 represents common drain terminals5..8+exposed drain. per actual drawing.



GPI08 is50Hz servo PWM; GPI09 is supply enable. BUF3 follows switched5V.

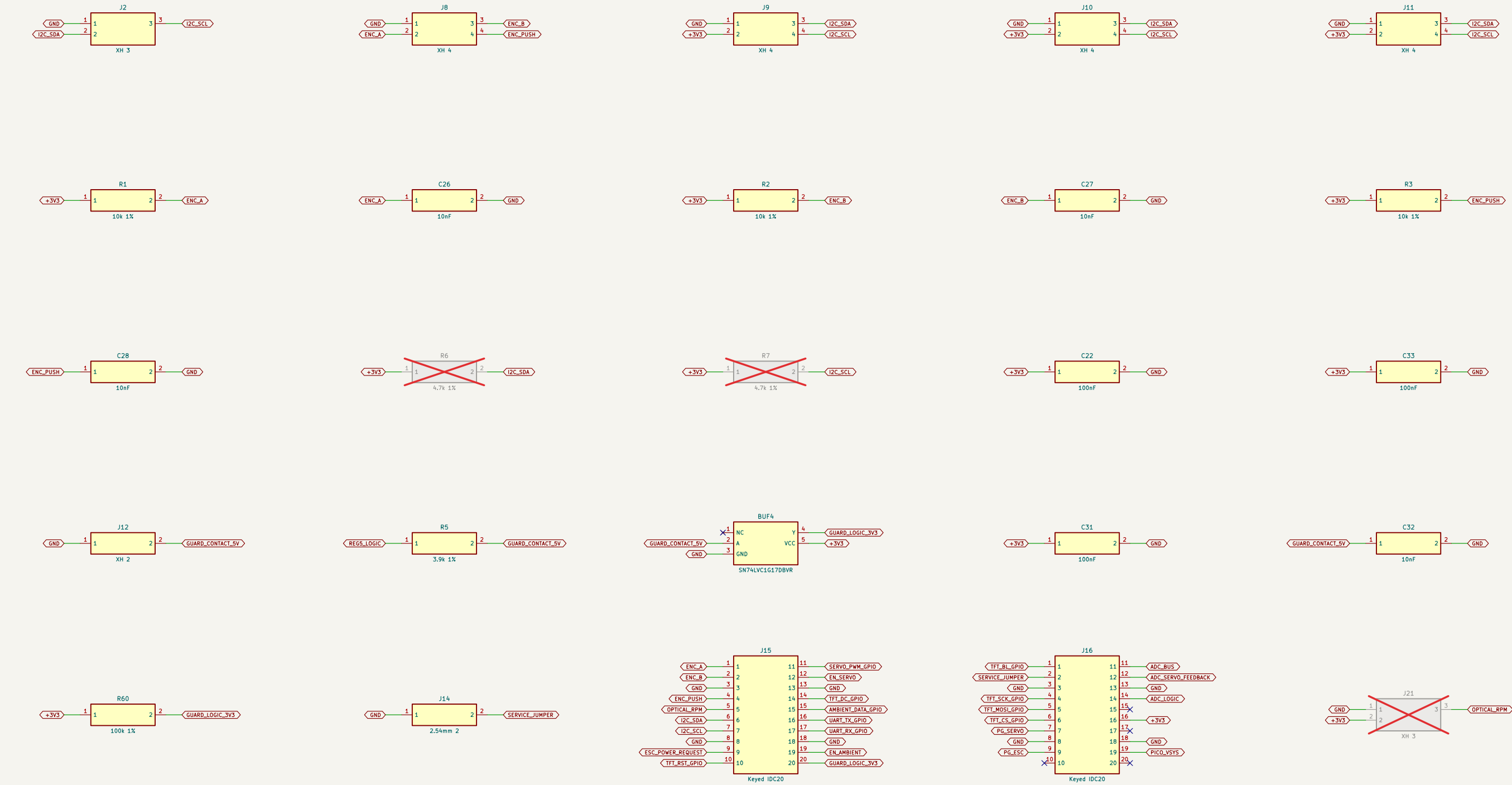
Timed movement and feedback checks bound jam energy; F3 does not replace jam cutoff.

Use real supplied20T servo horn; measured feedback table and end stops required.



Eight ambient pixels only. Screen replaces the front speed/power light bars.

BUF1 powered from switched ambient5V; its unused channels disabled and grounded.

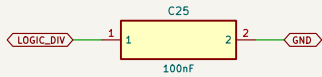
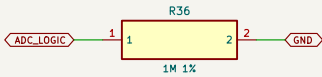
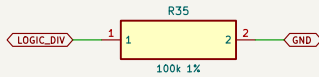
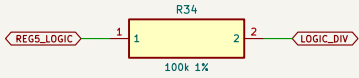
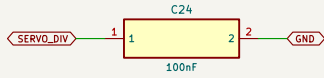
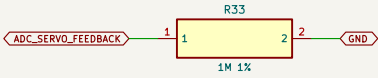
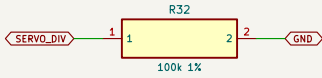
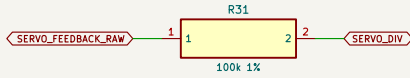
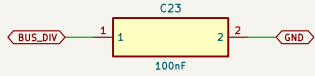
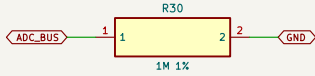
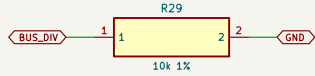
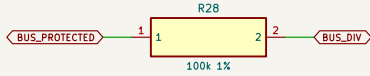
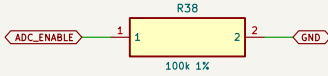
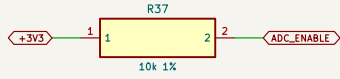
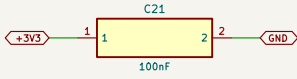
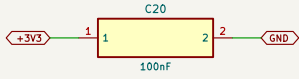
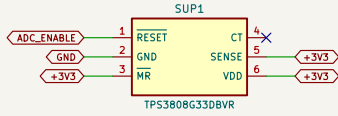
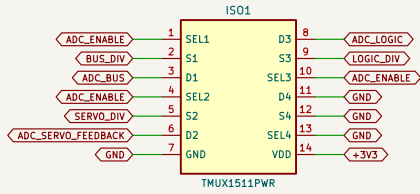


Guard COM/NO pulls rawguard LOW when grille is present; BUF4 translates it to GP15.

J15/J16 are keyed2x10 IDC headers, all20 contacts retained. OLD E3 keyed cable is incompatible.

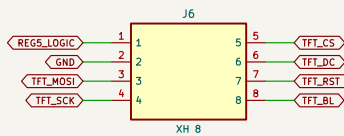
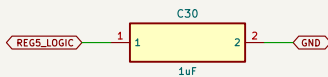
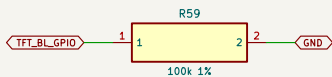
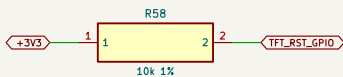
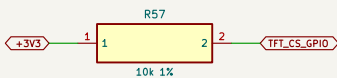
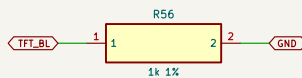
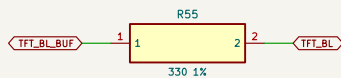
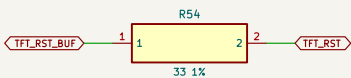
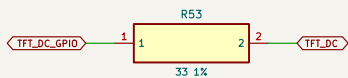
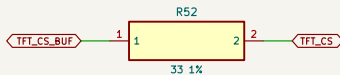
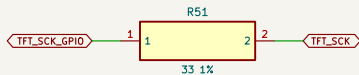
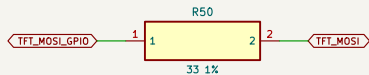
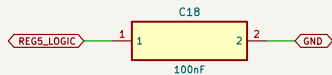
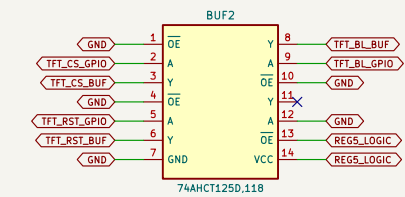
J15 pin n maps Pico physical n; J16 pin n maps physical n+20. Verify ribbon pin1 continuity.

R6/R7 DNP until aggregate breakout I2C pullups are measured. J21 optical RPM is DNP.



Filters precede TMUX1511. ADC-side1M pulldowns remain after isolation.

SUP1 enables ADC only after3.3V is healthy. Bus/logic scales11.1 and2.1 require calibration.



BUF2 uses unswitched5V. CS/RST pullups never connect5V directly to Pico.

C30/R56 fit at screen end. Top64rows reserve status band; wheel phase/countdown remain visible.